Yuen Tsan Hoi Zulniro (Brandon)

+61 423 958 406 | LinkedIn | Email | Portfolio | Open to relocation: QLD · VIC · WA (Regional)

CAREER PROFILE

UNSW BE (**Honours**) Aerospace Engineering graduate with professional experience across aviation, manufacturing, and construction environments. Naturally **curious**, **adaptable, and proactive**, I **enjoy exploring** new technologies, system, and processes. **Passionate** and **Motivated** by **continuous learning** and **improvement**. I **value hands-on experience** and enjoy understanding how designs, modifications, and systems operate before and after implementation. I **thrive** in **multidisciplinary** environments that **challenge** me to **develop** new skills, **solve** complex problems, and **collaborate** with people from **diverse** technical backgrounds. Seeking opportunities to broaden my technical knowledge, contribute to practical engineering solutions while continuing to grow into a versatile and **well-rounded engineer**. Eligible to work in Australia without restriction. Here is my personal (**Website Portfolio**)

PROFESSIONAL EXPERIENCE

NORITSU KOKI Co. Ltd.（ノーリツ鋼機株式会社）| Sydney, NSW — July 2026 – Present

Field Service Engineer | *Permanent Position*

- Diagnosed mechanical, electrical and software faults on using systematic **root cause analysis (RCA)**.
- Performed preventative maintenance to **maximize equipment reliability** and **operational performance**.
- Installed, commissioned and calibrated commercial imaging systems to operational standards.
- Analyzed equipment performance and implemented corrective actions to **improve system efficiency**.
- Interpreted **technical manuals**, **wiring diagrams** and **engineering documentation** to support fault diagnosis.
- Managed **customer service** activities while coordinate repairs, spare parts and maintenance schedules,
- Produced detailed technical reports and communicated engineering solutions with customers and internal teams.
- Assisted with **company relocation**, including physical labor, stock take, documentation, and logistics coordination.
- **Supported IT infrastructure** setup, including **LAN cabling**, **network connectivity**, and **workstation installation**.
- **Adapted quickly** to multidisciplinary tasks, demonstrating **initiative** and a **willingness to learn** new technical processes.

Qantas Airways | Sydney, NSW — Sep 2025 – April 2026

Cabin Engineer (Design & Drafting) | *Fixed Term Contract*

- Produced **CAD** drawings using **Draftsight** & **SolidWorks** for cabin reconfigurations and aircraft modifications.
- Managed **concurrent** design documentation workflows, version, **PDM** & **BOM** updates to maintain traceability.
- Collaborated with **cross-functional teams** to **resolve design conflicts** and **support** efficient modification delivery.
- Supported aircraft modification projects by maintaining accurate **technical records** throughout **asset lifecycle**.
- **Reviewed** & **translated** technical requirements into **production-ready** drawings and installation documentation.
- Applied CASA/EASA-compliant revision control, drawing verification, and document management processes to **ensure** engineering **accuracy** and **minimise** implementation **risks**.
- Contributed to **continuous improvement** of engineering documentation processes with (Templates|Excel Macros)

Metwest Steel Pty Ltd. | Perth, WA — Jun 2024 – Sept 2024

Assistant Engineer (CAD Operator) | *Structured 3-Month Internship*

- **Produced** fabrication drawings using **AutoCAD, SolidWorks, Tekla** to support manufacturing requirements.
- Applied **GD&T principles** to develop accurate, production-ready technical documentation.
- Utilized **Excel** & **Excel Macros** for documentation, track and improve **production** and **material usage efficiency**.
- Evaluated **20+** resin printing parameters through **RCA** to **identify** controlling variables to **optimise** 3D printing **quality** & **efficiency** for the display models.
- Developed and modified **Tekla Structures models** using engineering, architectural, and structural drawings.
- Coordinated between **engineers**, **fabrication** team, and **site contractors** to **resolve** design **discrepancies**, production **inefficiencies** to improve workflow efficiency.
- Conducted **site measurements** and incorporated fabrication and installation constraints into design updates.
- Gained **hands-on** exposure to fabrication processes, strengthened understanding of manufacturability.

TECHNICAL SKILLS

- **Engineering Tools:** *Proficient* in Draftsight, SolidWorks, AutoCAD, Inventor | Tekla Structures, CatiaV5, Rhino
- **Data Analysis & Simulation**: ANSYS CFD & FEA, MATLAB, OpenRocket, Xfoil, XFLR5, Flow5
- **Systems & Compliance**: Product Data management (**PDM**), Bill of materials (**BOM**) control, CASA/EASA
- **Productivity:** MS Excel Macros, Word, PowerPoint, Teams, Outlook, Photoshop, Acrobat

EDUCATION

University of New South Wales (UNSW) | Sydney, NSW Feb 2021 – May 2025
Bachelor of Engineering (Honours) in Aerospace Engineering
Class 2:1 (**Distinction WAM)** · UNSW Rocketry Team – (**LPR** & **MPR** development via **OpenRocket**)
Certificate of Merit for 2019 Future Pilot Program UNSW School of Aviation May 2020

KEY ENGINEERING PROJECTS

Research Thesis | Supersonic Bluff Body Wake Dynamics Dec 2024 – May 2025
- **Developed** high-fidelity CFD simulations of **Mach 3** bluff-body flows using compressible Navier-Stokes solvers with 2+ structured mesh refinement strategies and grid convergence verification.
- **Assessed** turbulence modelling approaches (**k-ε, k-ω SST, DES**) for separated supersonic wake flows.
- **Performed** spectral and statistical analysis of unsteady flow behavior using **MATLAB post-processing tools**.
- **Compared DES and k-ω SST** turbulence models, achieving a **34.5%** reduction in predicted wake thickness using **DES**.
- Investigated **unsteady flow characteristics** to identify key mechanisms influencing supersonic acoustic resonance.
- **Evaluated** numerical results against literature and experimental benchmarks to establish model validity, characterizing wake recirculation, shock structures, shear layers and reattachment phenomena.

Aeronautical Design Project | Kit-Plane Team Aircraft Design Sep 2024 – Dec 2024
- **Conducted** 2D/3D lift, drag and flight characteristic simulations using **Xfoil, XFLR5, Flow5**, and **MATLAB** to verify strict compliance with **CASA** and FAA Part 23 regulations.
- Executed **design** & **stability analysis** for aircraft weight & balance, fuselage, and internal cabin layouts.
- Evaluated **human anthropometric data** and **ergonomic principles** to optimize internal cockpit safety layouts.

Thermal Dynamics Project | Crossflow Heat Exchanger Optimization Feb 2024 – May 2024
- **Engineered** and **optimised** a crossflow HX through parametric CFD studies and MATLAB-automated performance tracking, achieving maximum design efficiency evaluated through computational validation.
- **Developed** a linear correlation data compiler within **CFD** to predict and map complex thermal performance.
- **Led** a team of 6 in heat exchanger optimization, task coordination, oversight of technical analysis and evaluations

Design Project | Crop Management Drone May 2023 – Aug 2023
In a group of 7, the objective of this project was to reduce physical presence and manual oversight on farms.
- **Managed** the technical functionality, control systems integration, and ergonomic design factors for an autonomous agricultural drone concept.
- **Economically** and innovatively to design our drone concept for farm management, integrating sensing & control systems for autonomous operations.
- **Aligned** drone design criteria with **CASA** and **EASA** unmanned aerial systems regulations to ensure compliance.

LANGUAGES & INTERESTS

- Music Accomplishments: Grade 8 (**Drum Kit Level 3 Trinity College London**) 16 Yrs| Piano Grade 6 (ABRSM)
- Outdoor Interests: Mountain & Road Biking, Photography, Media Editing, Camping, Hiking, Stargazing
- Languages: English (Fluent), Chinese (**Cantonese** – Native | **Mandarin** – Proficient)

CERTIFICATIONS & LICENSES

- Construction Induction White Card May 2026
- Driver's License July 2026

AWARDS

Recipient: Attained the Bronze Standard of the Hong Kong Award for Young People.	June 2019
Recipient: Won multiple awards in inter school swimming competition; representing YHKCC	2014 – 2020

References available upon request